DoggyFeed User Manual

Nov 3, 2026

1. Introduction

DoggyFeed plans what your dogs eat: it works out each dog's daily energy need, fills each bowl to that target from your own food library, and keeps a record of what was served. It runs in a web browser on a phone, tablet or computer.

What you can do with it

- Keep a profile for every dog: weight, desired weight, life stage, activity, meals per day and photo.
- Keep a library of foods with energy, protein, fat and fiber per 100 g, portions, stock, pictures and shop links.
- Fill each dog's bowl in the Meal planner and see the totals against the dog's energy target.
- Save bowls as reusable plans, record served bowls and look back at them in Bowls history.
- Keep a calendar of vet visits, grooming and other events, with reminders.
- Share one dog house with family members, with full or view-only (guest) access.

How your data is stored

All data belongs to a **dog house**: a household with its own dogs, foods, bowls, plans, calendar and users. Everything is saved on the DoggyFeed server in a MariaDB database, so every user of the dog house sees the same data on every device. Users of one dog house never see another dog house's data.

Languages

The app is available in English, Czech (Čeština) and Polish (Polski). Choose the language at the top right of the page or on the sign-in screen. Dates and times follow the chosen language. Names and other data you type in are never translated.

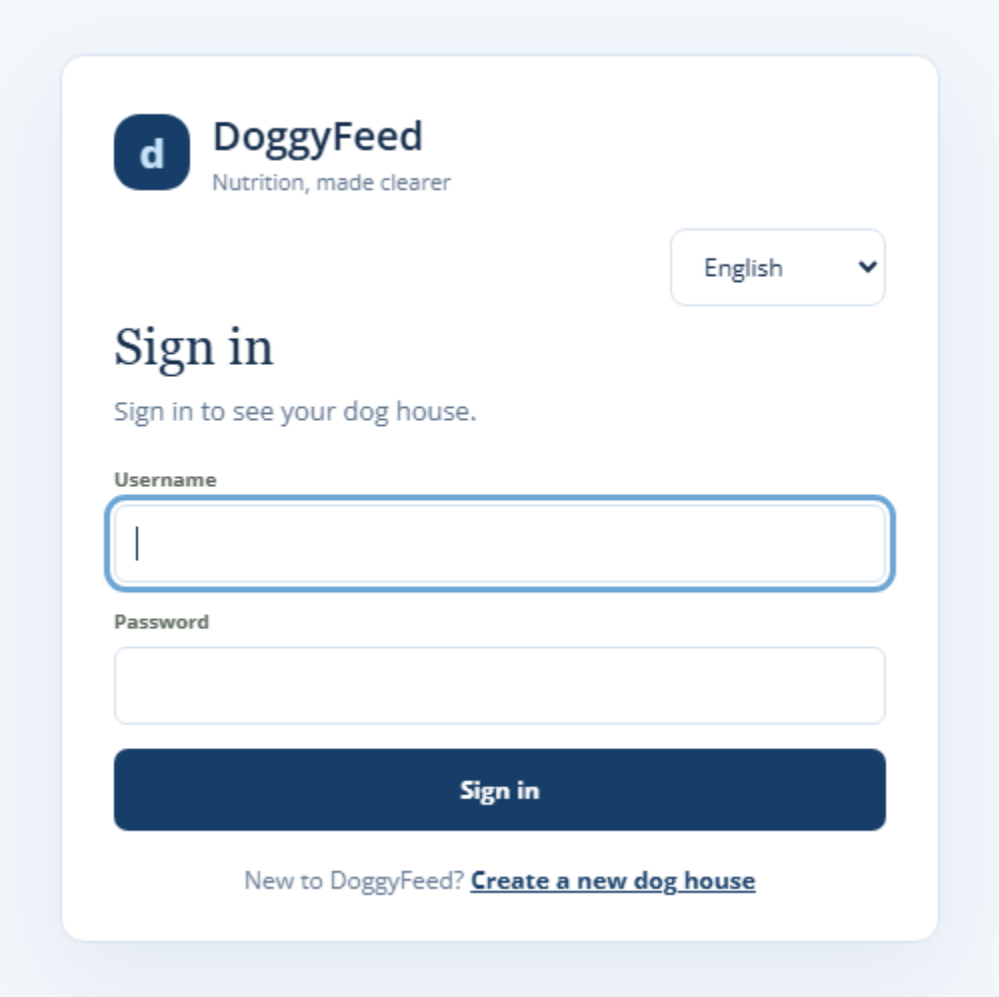
> *DoggyFeed is a planning aid, not veterinary advice. The energy targets are rough estimates, and the sample foods are examples: check nutrition values against the package or a trusted database.*

2. Getting started

You need a user in a dog house to use DoggyFeed. Either someone adds you to their dog house, or you create a new dog house yourself.

2.1 Sign in

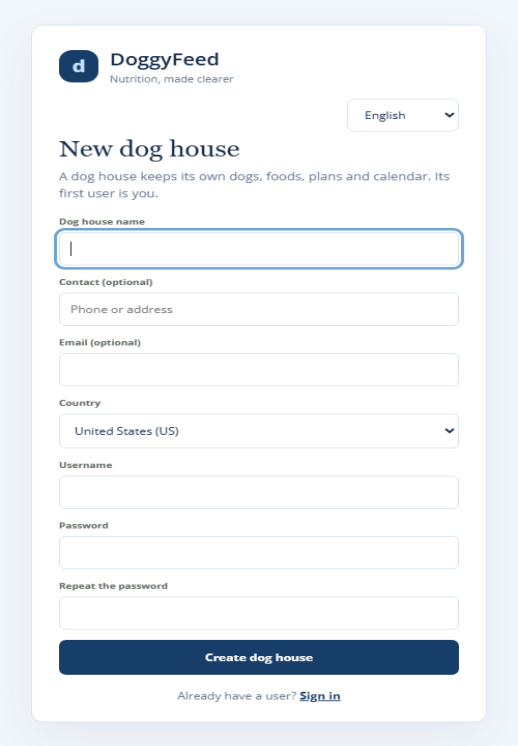
Open the DoggyFeed address in your browser. Enter your **Username** and **Password** and press **Sign in**. The language picker above the form switches the page language before you sign in.



A session lasts 30 days from its last use, so you stay signed in on your own devices. Use **Sign out** in the menu on shared computers.

2.2 Create a new dog house

On the sign-in screen, choose **Create a new dog house**. Fill in the form and press **Create dog house**. You become the first user of the new dog house.

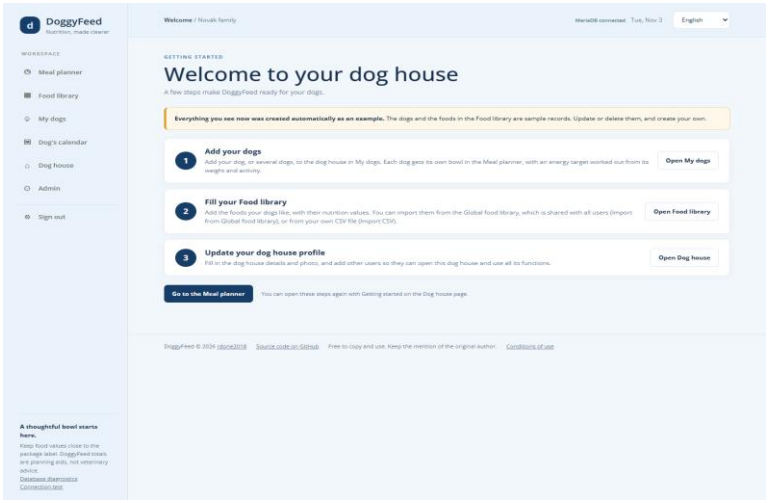


Field	Required	Example	Notes
Dog house name	yes	Novák family	Shown at the top of every page
Contact	no	+420 777 123 456	Phone or address
Email	no	family@example.com	
Country	yes	Czechia (CZ)	Pre-filled from your browser's language; used for food searches and imports
Username	yes	radek	Unique across all dog houses
Password / Repeat the password	yes	at least 8 characters	Stored only as a secure hash

The form is protected against automated sign-ups. If you send it within a few seconds of opening it, or try many times in a row, it asks you to wait and try again.

2.3 The Welcome page

A new dog house starts with **sample dogs and foods**, so you can try every function at once. The first time it opens, DoggyFeed shows the Welcome page with three setup steps.



1. **Add your dogs** in My dogs. Each dog gets its own bowl in the Meal planner, with an energy target worked out from its weight and activity.
2. **Fill your Food library** with the foods your dogs eat. Import them from the shared Global food library, from a CSV file, or search the web.
3. **Update your dog house profile**: details, photo, and other users who should share this dog house.

Update or delete the sample records, then press **Go to the Meal planner**. You can open the Welcome page again any time with **Getting started** on the Dog house page.

2.4 Roles

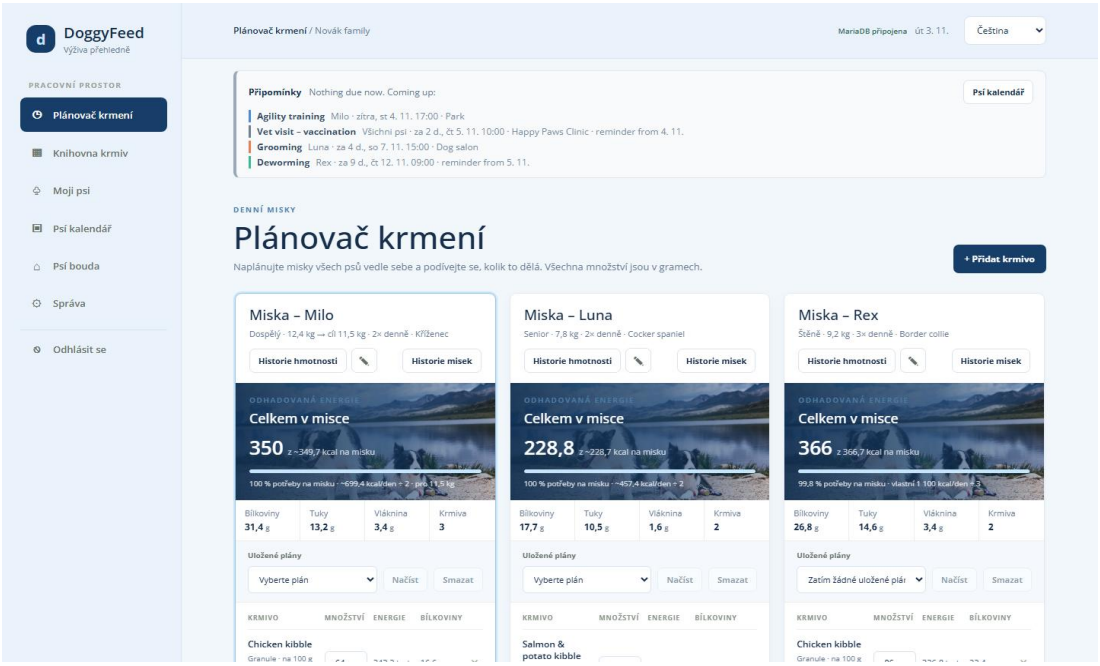
Every user belongs to exactly one dog house and has one of three roles.

Role	Can see	Can change	Typical user
User	Everything in the dog house	Everything in the dog house, including its users	Family members who feed the dogs
Guest	Everything in the dog house	Only their own password	A dog sitter or relative who only looks
Administrator	Everything in the dog house, plus the Admin console	Their dog house, plus all dog houses, users, the Global food library and countries	Whoever runs the DoggyFeed server

A guest sees the badge **Guest · view only** at the top of the page. Buttons that change data are hidden, and amounts and stock fields are read-only.

3. Screen layout

Every page has the same frame: the menu on the left, a top bar, the page content and a footer. On a phone the menu becomes a row of tabs at the top of the page.



Part	What it shows
Menu (left)	Meal planner, Food library, My dogs, Dog's calendar, Dog house, Admin (administrators only) and Sign out
Top bar	The page name and dog house name, the connection status, today's date and the language picker
Connection status	*MariaDB connected* when saves reach the server; *Saving…* while a save runs; *Browser storage only* or *Database save failed* when the server cannot be reached
Reminders ribbon	Events that are due or start today, then the next upcoming events (Meal planner and My dogs only)
Messages	Short confirmations at the bottom of the screen, such as "Food saved to database"
Footer	Copyright, a link to the source code on GitHub and the conditions of use
idebar note (administrators)	Links to Database diagnostics and the Connection test

Saving. Every change is saved to the server automatically; there is no Save button for the page as a whole. While a page is idle, it checks for changes made in another window or device every 30 seconds and loads them.

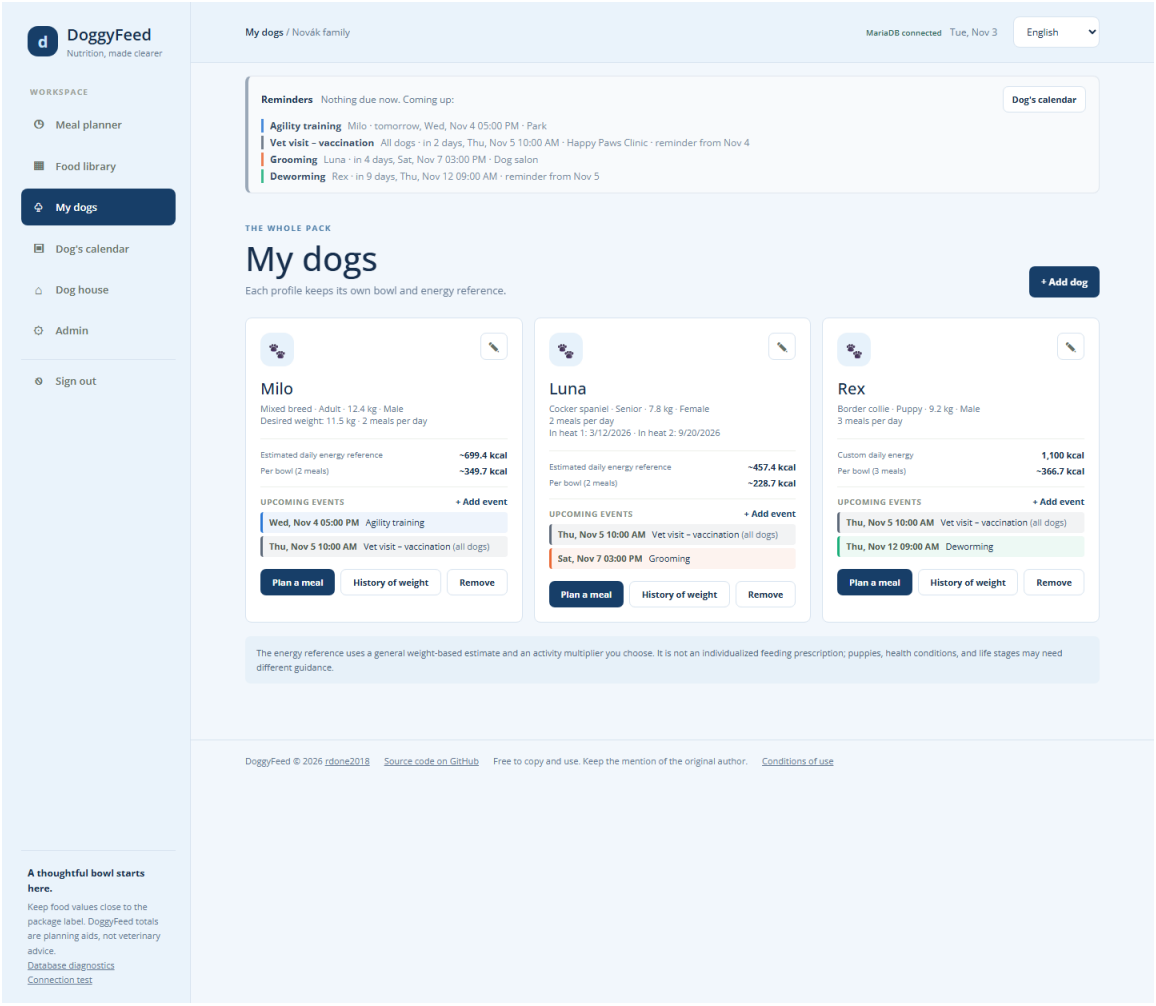
Banners. Two banners can appear at the top of a page:

- **Changes were made somewhere else**: the same dog house was changed on another device at nearly the same moment. Choose **Load the latest data** to see the newest data, or **Save my version anyway** to keep yours.
- **Restart the DoggyFeed server**: the server was updated but not restarted. Ask whoever runs the server to restart it.

4. Meal planner

5. My dogs

My dogs holds one card per dog with its details, its daily energy target and its next events. Each dog you add gets its own bowl in the Meal planner.

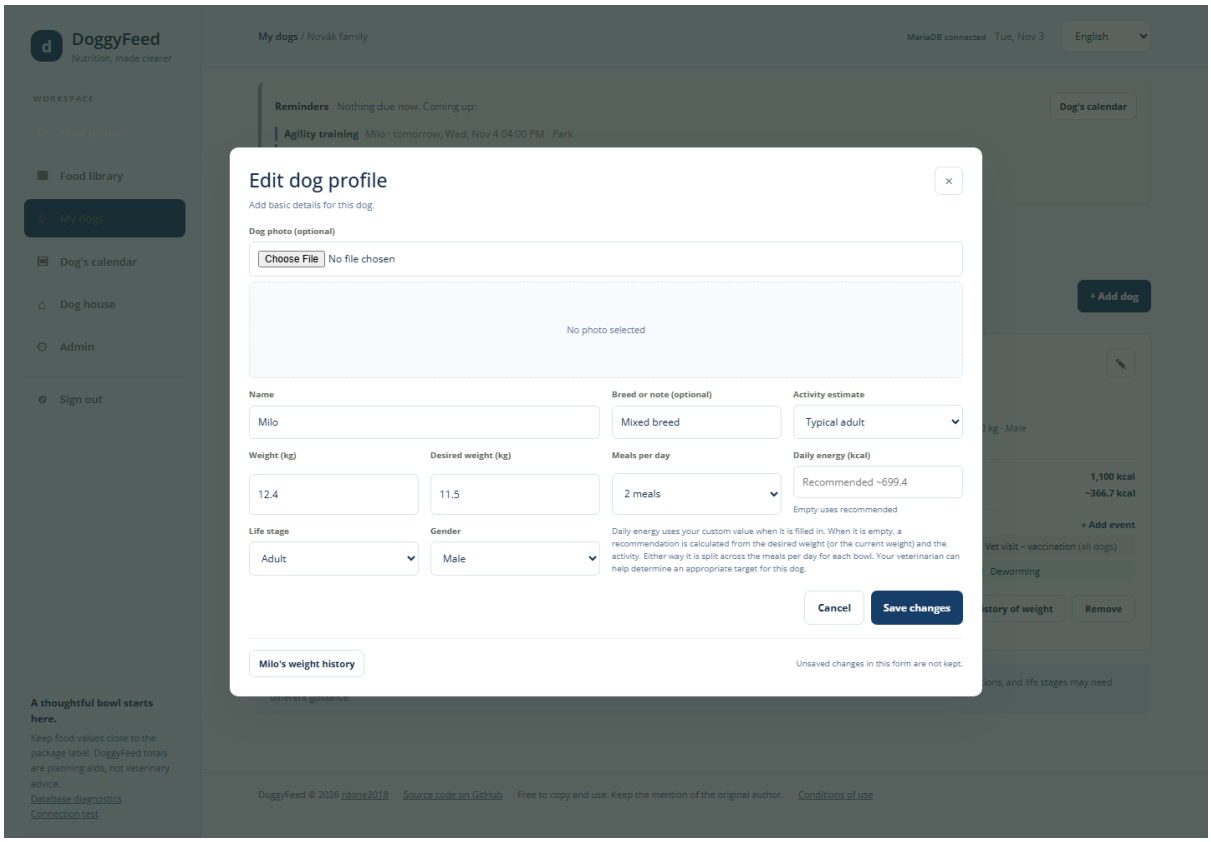


Each card shows the dog's photo, breed, life stage, weight, gender and meals per day, then:

- **Estimated daily energy reference** (or **Custom daily energy**) and the energy **Per bowl**.
- **Upcoming events**: the next three events within 60 days, for this dog and for all dogs. **+ Add event** adds one for this dog.
- **Plan a meal** opens the Meal planner with this dog selected; **History of weight** opens the weight history; **Remove** deletes the dog.

5.1 Adding or editing a dog

Press **+ Add dog**, or the pencil on a dog's card. Fill in the form and press **Save** (or **Save changes**).



Field	Values	Example	What it does
Dog photo	JPG, PNG or WebP, up to 8 MB		Shown on the dog's card and behind its bowl total; shrunk before upload
Name	up to 40 characters	Milo	Names the bowl, plans and events
Weight (kg)	0.1 to 250	12.4	Current weight; every change is kept in the weight history
Desired weight (kg)	0.1 to 250, optional	11.5	Target weight; when set, the energy target uses it instead of the weight
Life stage	Puppy, Adult, Senior	Adult	Shown on the card and bowl
Gender	Not specified, Female, Male	Male	For a female, two fields **In heat 1** and **In heat 2** appear for heat dates

Field	Values	Example	What it does
Meals per day	1 to 6	2 meals	The daily energy is split across this many bowls
Daily energy (kcal)	1 to 20,000, optional	empty	A custom daily target; empty uses the recommended estimate
Breed or note	up to 50 characters	Mixed breed	Free text
Activity estimate	see below	Typical adult	Multiplier in the energy estimate

5.2 How the energy target is worked out

When **Daily energy** is empty, DoggyFeed estimates it from the dog's weight and activity:

```
\text{Daily energy (kcal)} = 70 \times W^{0.75} \times \text{activity factor}
```

W is the desired weight in kg when set, otherwise the current weight. The activity factors are:

Activity estimate	Factor
Low activity	1.2
Gentle / senior	1.4
Typical adult	1.6
Active	1.8
Very active	2.0

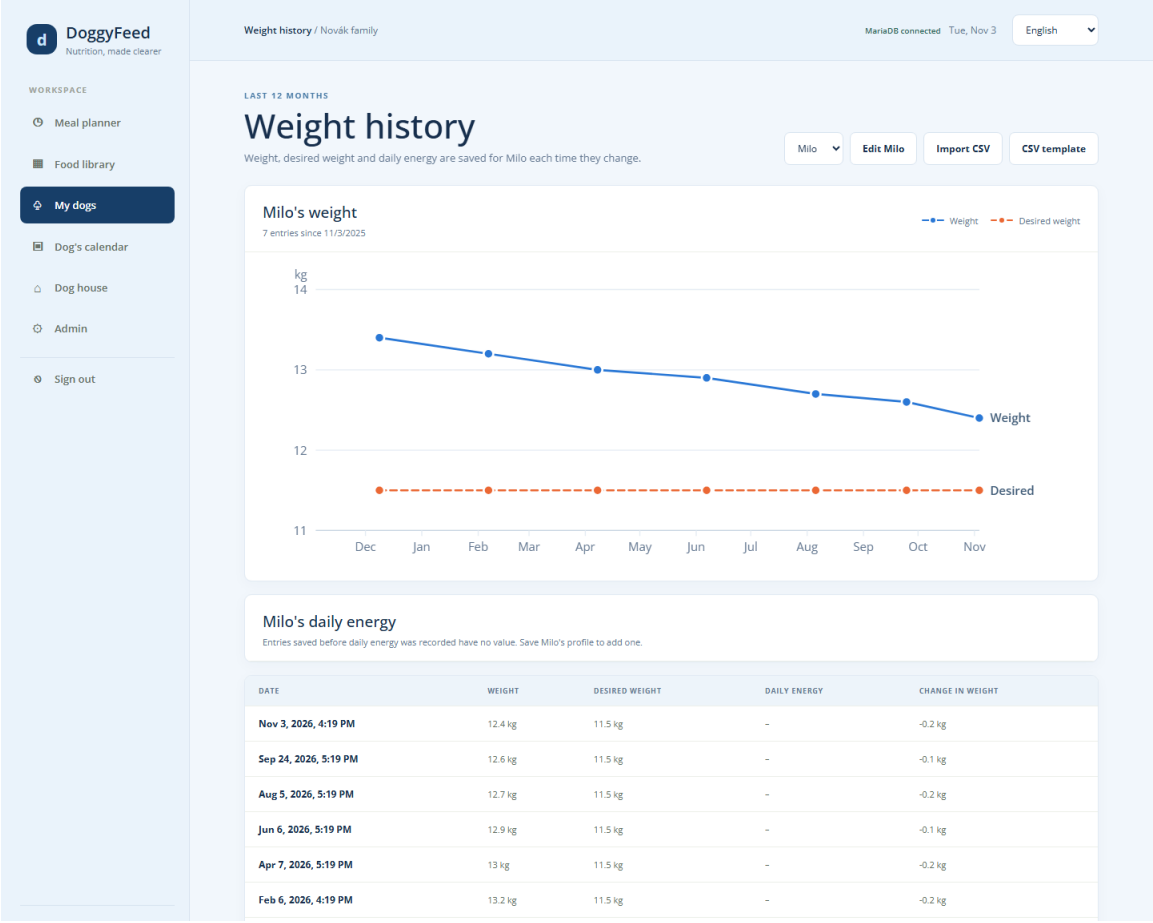
The target **per bowl** is the daily energy divided by the meals per day.

> ***Examples.*** *Milo, desired weight 11.5 kg, Typical adult: 70 × 11.5^0.75 × 1.6 = 699.4 kcal a day; with 2 meals, 349.7 kcal per bowl. Luna, 7.8 kg, Gentle / senior: 70 × 7.8^0.75 × 1.4 = 457.4 kcal a day; 228.7 kcal per bowl. Rex has a custom daily energy of 1,100 kcal and 3 meals: 366.7 kcal per bowl.*

The estimate is general and weight-based. Puppies, pregnant or nursing dogs and dogs with health conditions may need different amounts; ask your veterinarian.

5.3 Weight history

History of weight (on the dog's card, bowl or edit form) opens a chart of the dog's weight and desired weight over the last 12 months, a chart of its daily energy, and a table of every saved entry with the change in weight.

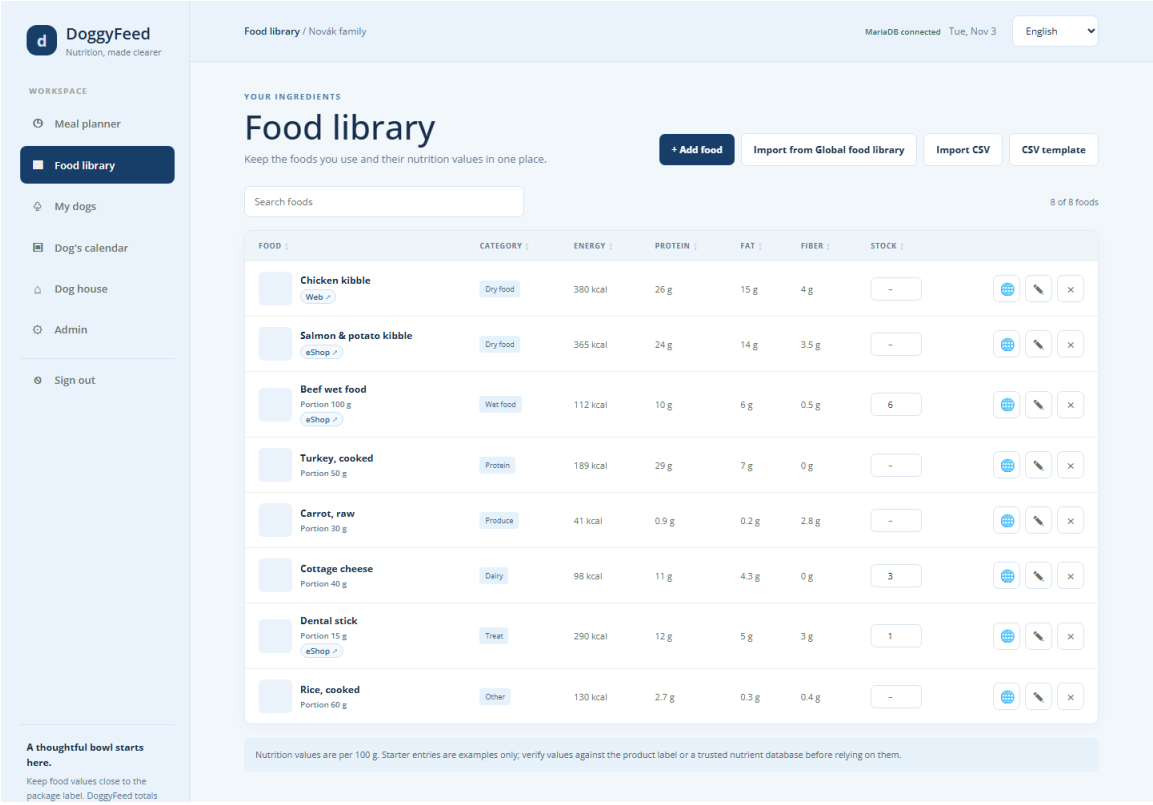


A new entry is saved every time you save a changed weight, desired weight or daily energy. The dog picker at the top switches between dogs, and **Edit** opens the dog's form.

Import CSV loads older history from a file; **CSV template** downloads an example. The file format is described in section 10.4. An imported row for a day that already has an entry replaces it.

6. Food library

The Food library lists every food your dog house uses, with its nutrition values per 100 g. The Meal planner only offers foods from this library.



6.1 The list

- **Search foods** filters the list by name or category as you type.
- Click a **column name** to sort by it: the first click sorts ascending (▲), the next descending (▼). Foods whose stock is not tracked stay at the end.
- **Web ↗** and **eShop ↗** under a food's name open its manufacturer's website or shop page in a new tab.
- Row buttons: 🌐 exports the food to the Global food library (section 6.6), ✎ edits it (or double-click the row), **×** deletes it.

6.2 Adding or editing a food

Press **+ Add food**, or ✎ on a food. Fill in the form and press **Save**.

Field	Required	Example	Notes
Food name	yes	Salmon & potato kibble	Up to 60 characters
Category	yes	Dry food	Dry food, Wet food, Protein, Produce, Dairy, Treat or Other
Food picture	no		JPG, PNG or WebP, up to 8 MB; shrunk before upload
Energy (kcal) per 100 g	yes	365	From the package (see section 10.3 for kcal/kg)
Protein, Fat, Fiber (g) per 100 g	yes	24 / 14 / 3.5	The "analytical constituents" on the package, in % = g per 100 g
Portion (g)	no	100	Amount added to a bowl at once; hidden for Dry food
Stock (pieces)	no	6	Empty = not tracked
Manufacturer website	no	https://www.example-petfood.com	Shown as **Web ↗**
eShop link	no	https://www.example-shop.com/salmon	Shown as **eShop ↗** and linked from out-of-stock events

Portion. A food with a portion is added to a bowl with that amount. A food without one (always Dry food) is auto-filled to the bowl's energy target (section 4.2).

6.3 Stock

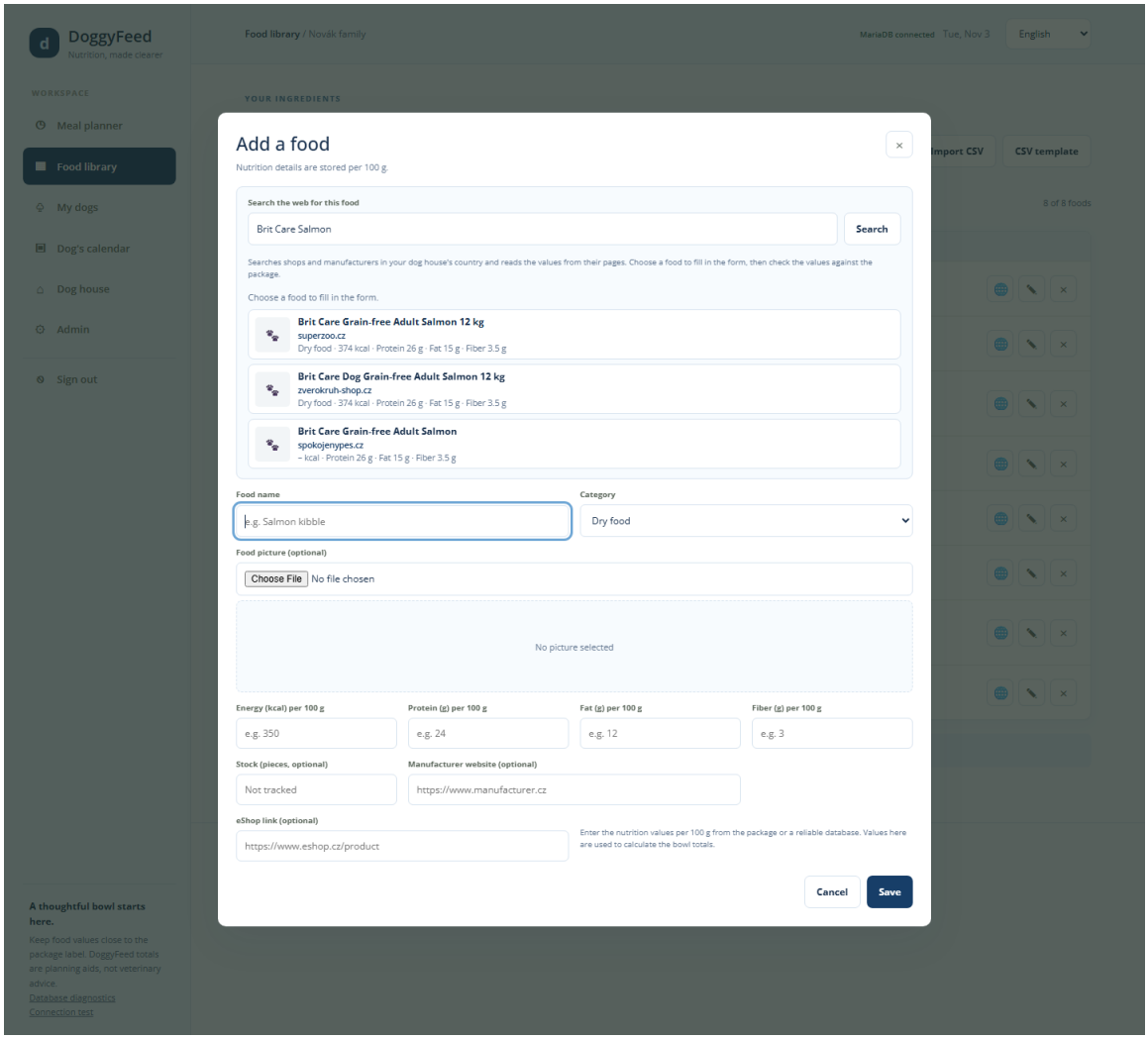
Stock counts **pieces**, such as cans, pouches or sticks. Type a new number in the **Stock** column; it is saved when you leave the field or press Enter (Escape puts the old value back). An empty field stops tracking.

- **Bowl served** takes the used pieces off every food that is not Dry food: grams served ÷ portion, or 1 piece for a food without a portion.
- A food at 0 is shown in red. Stock never goes below 0.
- When a food drops below 2 pieces, an event **"<food> - out of stock"** is added to the Dog's calendar for all dogs, linking to its eShop page. It disappears once the stock is 2 or more again.

> ***Example.*** *Beef wet food has a portion of 100 g and a stock of 6 cans. Luna is served a bowl with 100 g of it: the stock drops to 5.*

6.4 Searching the web for a food

The **Add a food** form starts with **Search the web for this food**. Type the brand and product name and press **Search**.



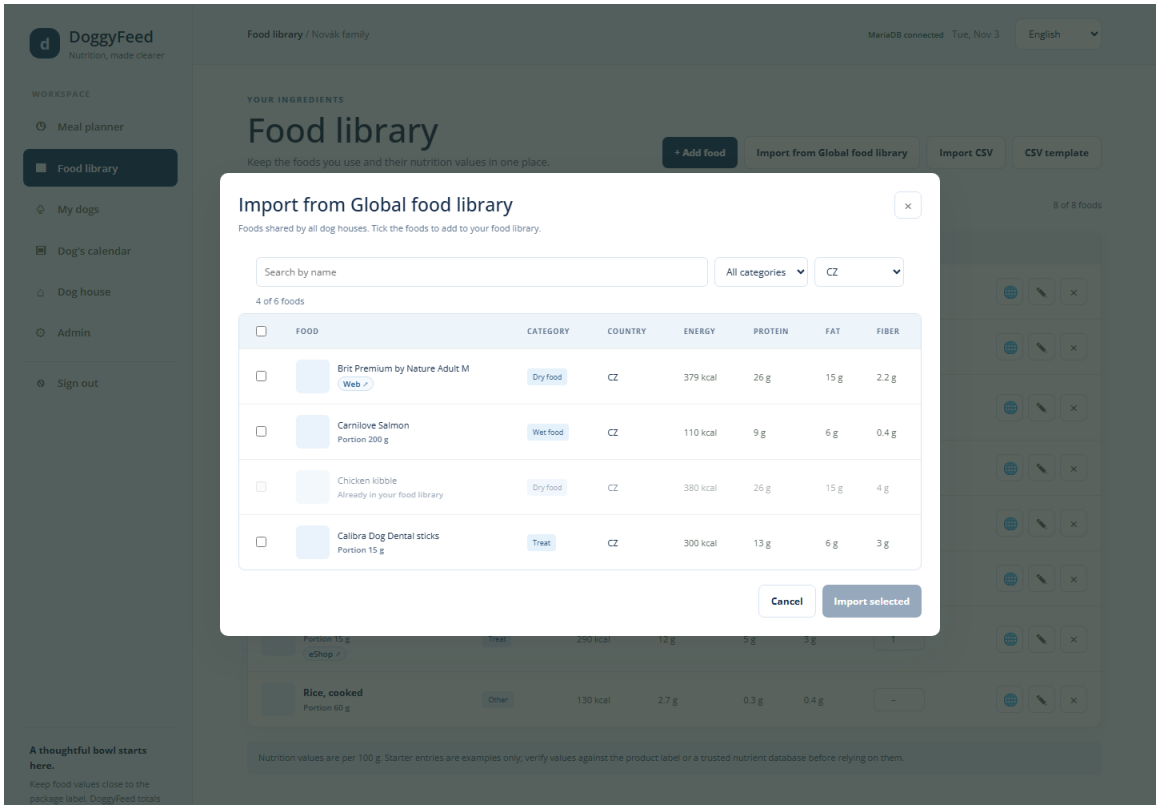
DoggyFeed searches shops and manufacturers in your dog house's country and language, and reads the values from the pages it finds. Each result shows the shop's site and the values it found; **–** means a value was not found. Results with more values come first.

Choosing a result fills in the name, category, energy, protein, fat, fiber, picture and eShop link. Fill in the empty values and **check every value against the package** before saving: shop pages can be wrong or out of date.

The web search needs a search key set up on the server (see section 11). Without it, results come from the Open Pet Food Facts database, and the form says so.

6.5 Importing foods

Import from Global food library lists foods shared by all dog houses, filtered by name, category and country. The country filter starts at your dog house's country. Tick the foods you want and press **Import selected**; their pictures come along. Foods already in your library are greyed out.



Import CSV loads foods from a CSV file; **CSV template** downloads an example. A row whose name matches a food in the library updates it and keeps its picture; other rows are added. The file format is in section 10.4.

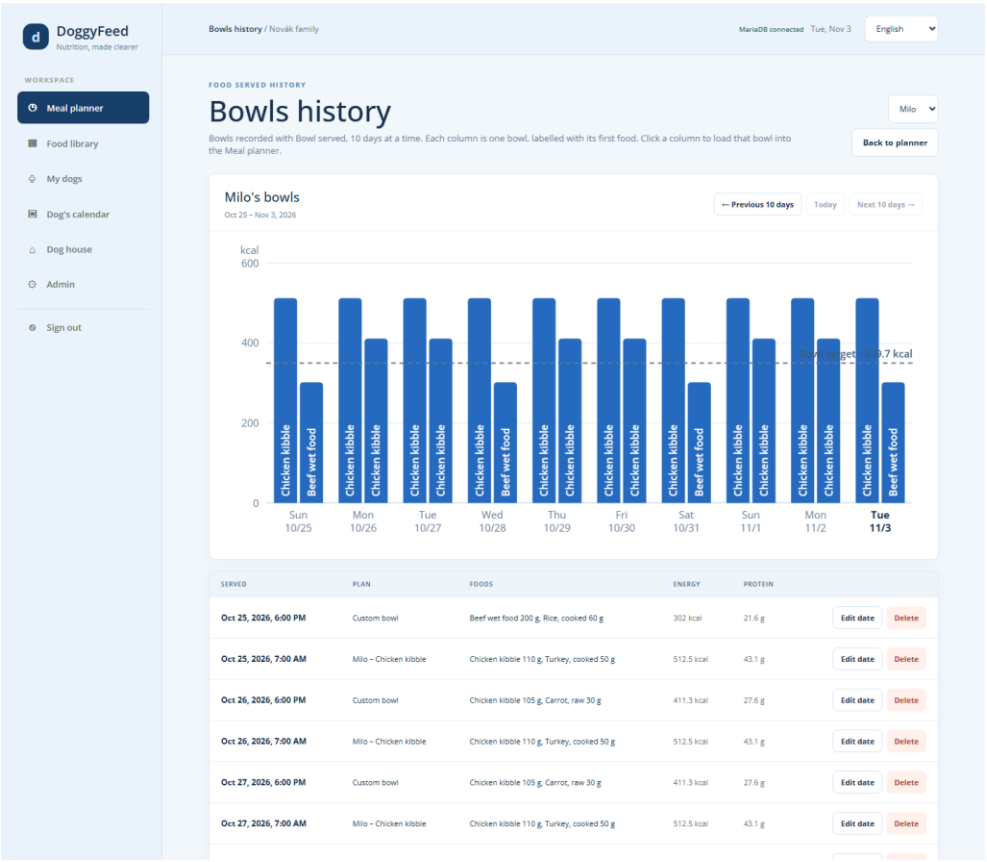
6.6 Sharing a food with everyone

🌐 on a food copies it, with its picture but without its portion, to the Global food library under your dog house's country. If a food with the same name and country is already there, administrators are asked whether to replace it; other users are told it already exists.

7. Bowls history and Dog's calendar

7.1 Bowls history

Bowls history shows every bowl recorded with **Bowl served**, 10 days at a time. Open it with **Bowls history** on a dog's bowl in the Meal planner.

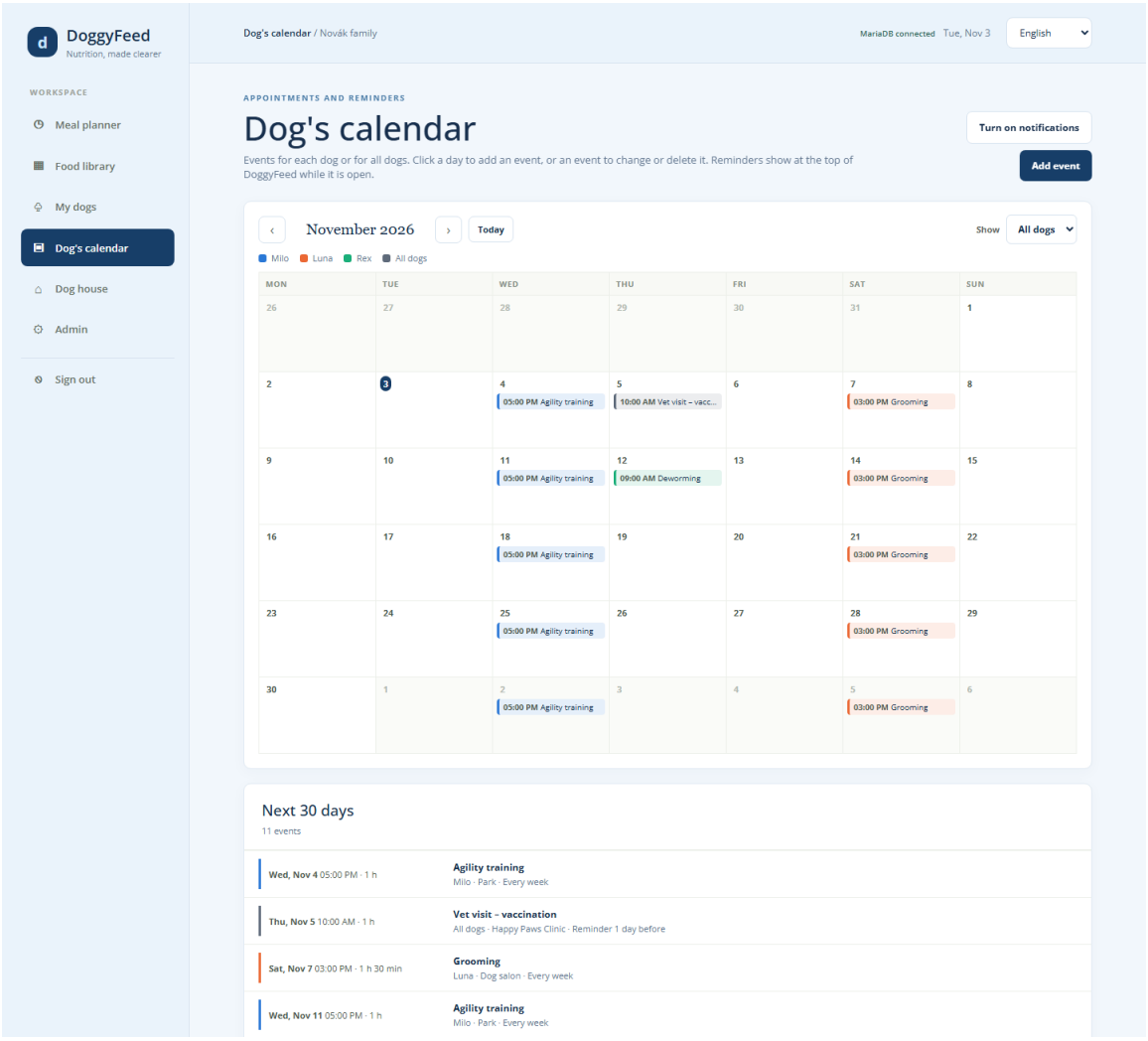


- **The chart** has one column per served bowl, labelled with its first food, and a dashed line at the dog's bowl target. A column well above the line means a bowl with more energy than planned.
- **Click a column** to load that bowl back into the Meal planner, for example to serve the same bowl again.
- The dog picker switches dogs; **Previous 10 days**, **Today** and **Next 10 days** move through time; **Back to planner** returns to the Meal planner.
- **The list** below shows each bowl's time, plan name, foods with amounts, energy and protein. **Edit date** moves a bowl to another date and time; **Delete** removes it. Deleting a bowl does not put its pieces back into stock.

> ***Example.*** *On Oct 25 Milo got two bowls: "Milo – Chicken kibble" at 7:00 AM (110 g kibble + 50 g turkey = 512.5 kcal) and a custom bowl at 6:00 PM (302 kcal). The morning bowl is 163 kcal over the 349.7 kcal target.*

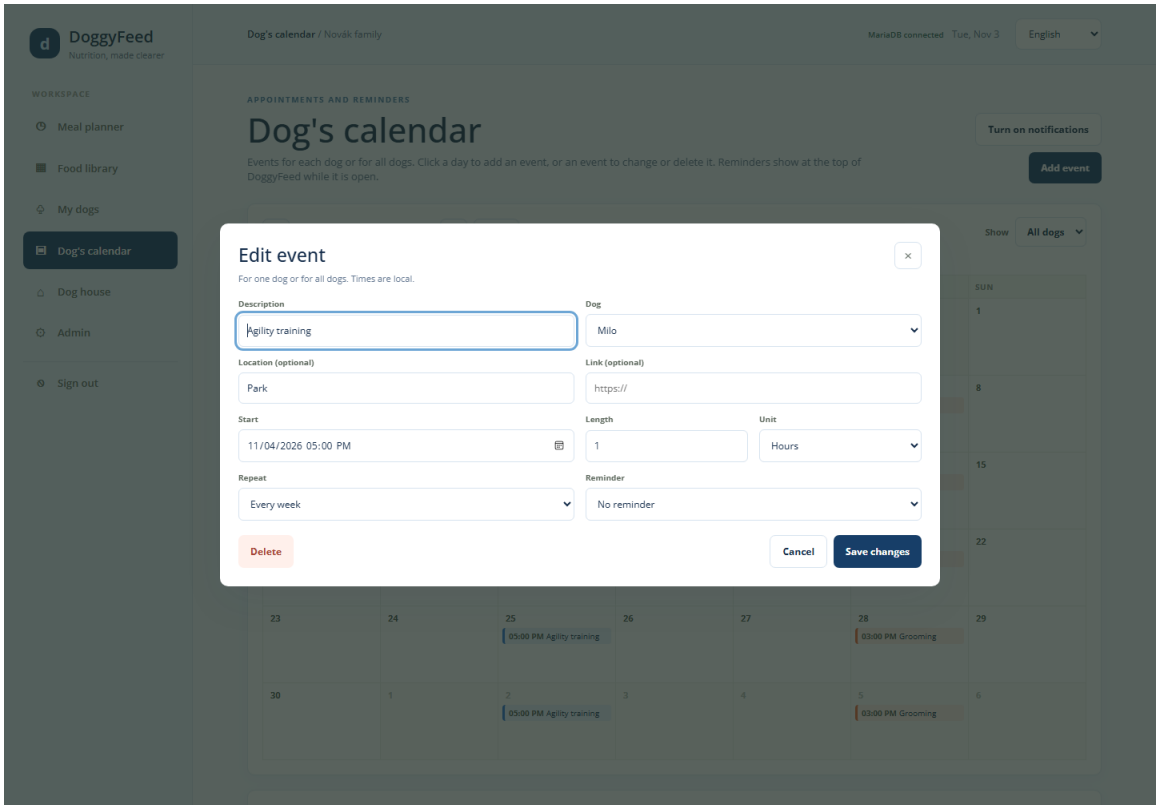
7.2 Dog's calendar

The Dog's calendar keeps appointments for one dog or for all dogs: vet visits, grooming, training, medication. It shows one month at a time and lists the next 30 days below.



- **‹** and **›** change the month; **Today** goes back to this month.
- **Show** filters the events by dog. Each dog has its own colour; grey is for events for all dogs.
- **Click a day** to add an event on that day, or press **Add event**.
- **Click an event** to change or delete it.
- **Turn on notifications** lets the browser show reminders as system notifications.

7.3 Events



Field	Values	Example
Description	text	Vet visit – vaccination
Dog	one dog, or All dogs	All dogs
Location	optional text	Happy Paws Clinic
Link	optional web address	The clinic's booking page
Start	date and time (local time)	Thu, Nov 5, 10:00 AM
Length and Unit	number with minutes, hours or days	1 hour
Repeat	Does not repeat, Every day, Every week, Every year, Every … days	Every week
Reminder	No reminder, 1 day, 1 week or 1 month before, … days before	1 day before

7.4 Reminders

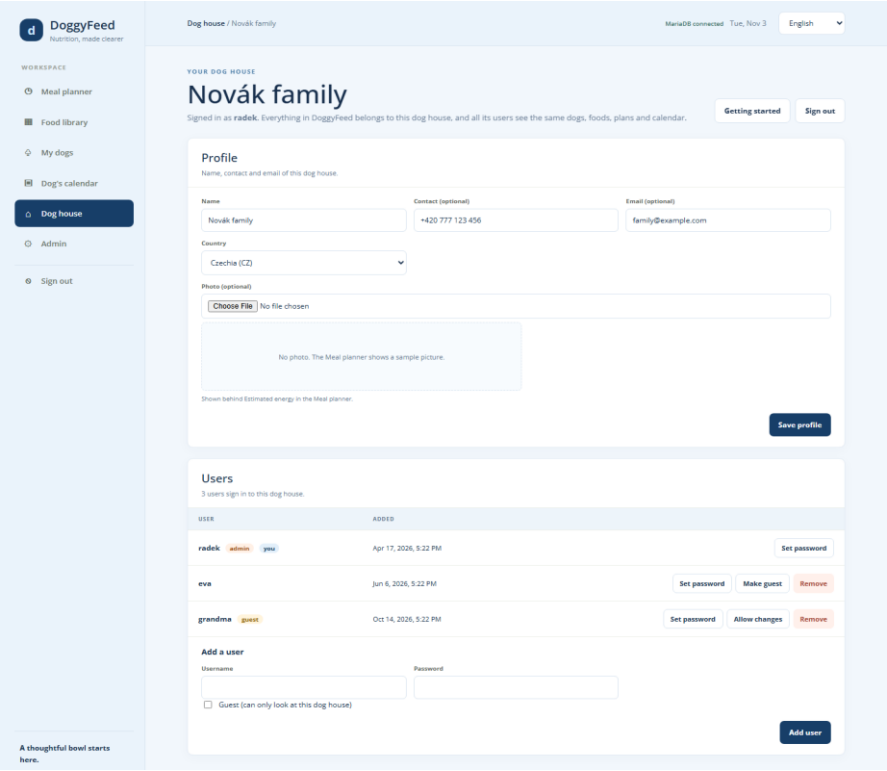
From the reminder time until the event starts, the event is listed in the **Reminders** ribbon at the top of the Meal planner and My dogs, with a button to dismiss it. Events that start today are always listed. Below what is due, the ribbon lists the next 6 upcoming events within 90 days.

When notifications are turned on, each reminder also appears once as a system notification. Reminders only appear while DoggyFeed is open, and notifications need DoggyFeed opened over https.

Out of stock. When a food's stock drops below 2 pieces, DoggyFeed adds the event "<food> - out of stock" for all dogs automatically (section 6.3).

8. Dog house

The Dog house page holds your dog house's profile and the users who share it. All its users see the same dogs, foods, plans and calendar.



8.1 Profile

Field	Example	Notes
Name	Novák family	Shown in the top bar
Contact	+420 777 123 456	Optional
Email	family@example.com	Optional
Country	Czechia (CZ)	Chosen from the list of countries; sets where the web food search looks and which country is pre-selected when importing from the Global food library
Photo		Optional; shown behind the bowl total of every dog without its own photo. Without any photo, a sample picture is shown

Press **Save profile** to save changes.

8.2 Users

The list shows every user of the dog house with their role (**admin**, **guest**, and **you** for yourself) and the date they were added.

- **Add a user**: type a username and password (at least 8 characters) and press **Add user**. Tick **Guest** for a user who should only look. Usernames are unique across all dog houses.
- **Set password** gives a user a new password. That user is signed out on all other devices.
- **Make guest** turns a user into a view-only guest; **Allow changes** turns a guest back into a normal user.
- **Remove** deletes the user.

Rules: nobody can change their own role or remove themselves, administrators cannot be guests, and every dog house keeps at least one user who is not a guest.

> ***Example.*** *To let a dog sitter check the feeding plan while you are away, add the user "sitter" with* ***Guest*** *ticked. They see every bowl, plan and event but cannot change anything. Remove them when you are back.*

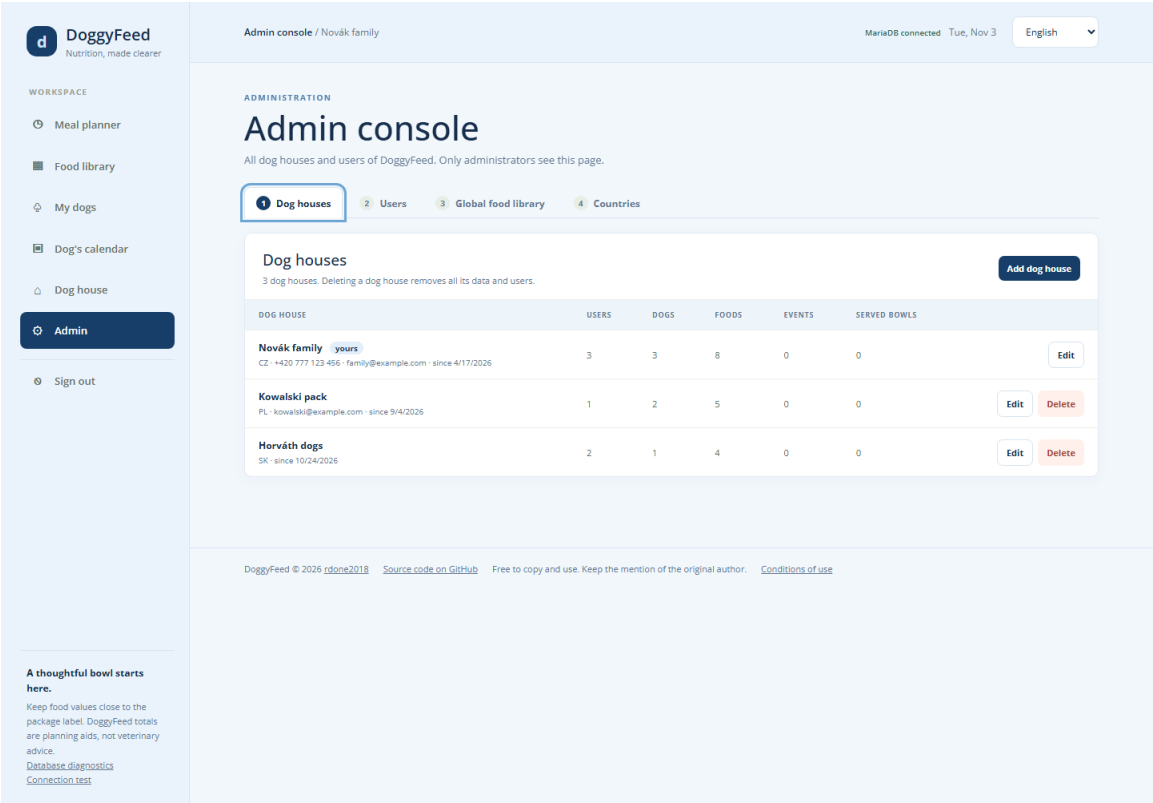
8.3 Other buttons

- **Getting started** opens the Welcome page with the setup steps (section 2.3).
- **Sign out** signs you out on this device.

9. Admin console (available only for Administrator)

The Admin console manages the whole DoggyFeed server: every dog house, every user, the shared Global food library and the list of countries. Only administrators see **Admin** in the menu. The page has four tabs, and it remembers the last one you opened.

9.1 Dog houses



The list shows each dog house with its country, contact, start date and counts of users, dogs, foods, events and served bowls.

- **Add dog house** creates an empty dog house.
- **Edit** changes its name, contact, email and country.
- **Delete** removes the dog house **with all its data and users**, after you type its name to confirm. You cannot delete your own dog house.

9.2 Users

[image: Admin console, Users tab]

- **Add user** adds a user to any dog house, optionally as an administrator or a guest.
- **Set password**, **Move** (to another dog house), **Make guest** / **Allow changes**, **Make admin** / **Remove admin** and **Delete**.
- You cannot delete or demote yourself, and DoggyFeed always keeps at least one administrator. The user who first set up DoggyFeed is the first administrator.

9.3 Global food library

[image: Admin console, Global food library tab]

The Global food library holds foods shared by all dog houses, each with a country. Users import from it into their own Food library (section 6.5).

- Filter by name, category and country.
- **Add food**, or double-click a food (or ✎) to edit it, with its picture. Portion is hidden for Dry food, as in the Food library.
- **×** deletes one food; tick several and press **Delete selected** to delete them together.
- **Import CSV** adds or updates up to 5,000 foods at a time (format in section 10.4). A row whose name and country match an existing food updates it and keeps its picture.
- **Import pictures** adds pictures to foods already in the list: choose JPG, PNG or WebP files named after the foods, for example `Chicken kibble.jpg`. Letter case does not matter. Files that match no food are listed and skipped. Import the CSV first, then the pictures.

9.4 Countries

[image: Admin console, Countries tab]

The countries table lists every country a dog house can be in: a two-letter code (ISO 3166-1, plus XK for Kosovo) and a name, with the number of dog houses in it. Search by code or name, **Add country**, edit (✎) or delete (**×**). Changing a code also changes it for the dog houses in that country. A country with dog houses in it cannot be deleted.

9.5 Diagnostics and Connection test

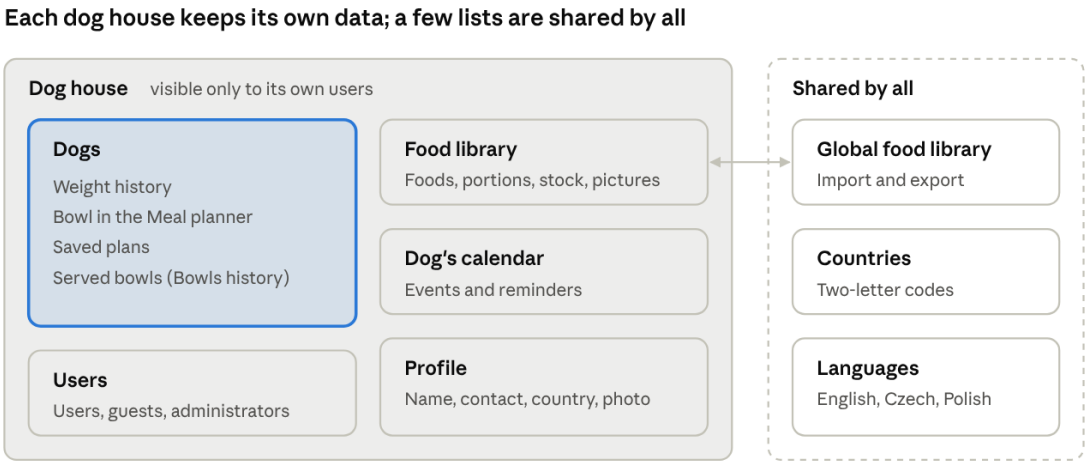
Administrators see two links at the bottom of the menu:

- **Database diagnostics** shows the database tables, row counts and stored values, and the server's error log. It only opens on the server computer itself (`http://127.0.0.1:3000/diagnostics`).
- **Connection test** sends test requests of different sizes to the server and reports which ones arrive. Run it through the address you normally use, such as your internet domain, when saves fail there but work locally. Nothing is saved.

10. Data explanation

10.1 What DoggyFeed stores

Everything except the Global food library and the countries belongs to one dog house and is visible only to its users.



what belongs to a dog house and what is shared

Record	Belongs to	Holds
Dog house	the server	Name, contact, email, country, photo
User	a dog house	Username, password (as a secure hash only), administrator and guest flags
Dog	a dog house	Name, weight, desired weight, life stage, gender, heat dates, meals per day, custom daily energy, activity, breed, photo
Weight history entry	a dog	Date and time, weight, desired weight, daily energy, whether the energy was custom
Food	a dog house	Name, category, kcal, protein, fat, fiber per 100 g, portion, stock, manufacturer and shop links, picture
Bowl (meal item)	a dog	Food, amount in grams, auto-fill on or off
Saved plan	a dog (or any dog)	Name, foods with amounts, saved and updated dates
Served bowl	a dog	Time served, plan name, target, each food's name, amount and values at that moment
Calendar event	a dog or all dogs	Description, location, link, start, length, repeat,

Record	Belongs to	Holds
		reminder
Global food	the server	Like a food, plus a country, without stock
Country	the server	Two-letter code and name
Language and translation	the server	Page texts in English, Czech and Polish

A served bowl keeps a copy of each food's values, so Bowls history stays correct even after you change or delete the food.

10.2 Units

Value	Unit
Weight, desired weight	kilograms (kg)
Energy of a food	kcal per 100 g
Protein, fat, fiber of a food	grams per 100 g (the same number as % on the package)
Amounts in a bowl, portion	grams (g)
Daily energy, bowl target	kcal
Stock	pieces (cans, pouches, sticks, bags)

10.3 Calculations

Bowl totals. For each food in the bowl, its values are scaled by the amount:

```
\text{energy of a food in the bowl} = \frac{\text{amount (g)} \times \text{kcal per 100 g}}{100}
```

Protein, fat and fiber are scaled the same way. The bowl total is the sum over all foods; the percentage is the bowl total divided by the bowl target.

> ***Example.*** *50 g of Turkey, cooked (189 kcal, 29 g protein per 100 g) gives 50 × 189 ÷ 100 = 94.5 kcal and 14.5 g protein.*

Daily energy and bowl target. See section 5.2: 70 × weight^0.75 × activity factor, or the custom daily energy, divided by the meals per day.

Reading a package. Pet food packages list values in different ways. Convert them before typing them in:

On the package	Type in DoggyFeed	Example
Energy in kcal/100 g	the same number	374 kcal/100 g → 374
Energy in kcal/kg	divide by 10	3,740 kcal/kg → 374
Energy in MJ/kg	multiply by 23.9	15.6 MJ/kg → 373
Crude protein, crude fat, crude fibre in %	the same number, in grams	Crude protein 26 % → 26

Stock. Each Bowl served takes off grams ÷ portion pieces for every food that is not Dry food, or 1 piece for a food without a portion.

10.4 CSV file formats

CSV files are plain text tables, one row per record, with a header row naming the columns. Comma, semicolon or tab separators are accepted, and decimal commas work. Every import page has a **CSV template** button that downloads the example below.

Food library (Import CSV). `name` and `kcal` are required; the other columns are optional. A row whose name matches a food updates it. Unknown categories become Other. An empty portion, stock or link clears that value; a file without the column keeps existing values.

```
name,category,kcal,protein,fat,fiber,portion,stock,manufacturer_url,shop_url
Salmon kibble,Dry
food,365,24,14,3.5,,,https://www.example.com,https://shop.example.com/salmon-kibble
Chicken in jelly pouch,Wet food,85,9,4.5,0.3,100,24,,
Carrot raw,Produce,41,0.9,0.2,2.8,40,,,
```

Global food library (administrators). Like the food library, plus `country` (a two-letter code) and without `stock`. A row whose name and country match a global food updates it. At most 5,000 rows per file.

```
name,category,country,kcal,protein,fat,fiber,portion,manufacturer_url,shop_url
Salmon kibble,Dry
food,CZ,365,24,14,3.5,,https://www.example.com,https://shop.example.com/salmon-kibble
Chicken in jelly pouch,Wet food,DE,85,9,4.5,0.3,100,,
```

Weight history. `date` and `weight` are required. Dates can be `YYYY-MM-DD` or `DD.MM.YYYY`, optionally with a time. `daily_energy_custom` is yes or no. A row for a day that already has an entry replaces it.

```
date,weight,desired_weight,daily_energy,daily_energy_custom
2026-01-15,12.4,11.5,,
2026-02-15,12.1,11.5,650,yes
2026-03-15 08:30,11.9,11.5,620,yes
```

11. Troubleshooting and FAQ

Problem	Cause	What to do
Banner "Changes were made somewhere else"	The dog house was changed on another device or tab at nearly the same moment	**Load the latest data** to see the newest version, or **Save my version anyway** to keep yours. Close DoggyFeed in tabs you are not using.
Status "Browser storage only" or "Database save failed"	The page cannot reach the DoggyFeed server	Check your internet connection and reload. Changes made meanwhile are kept only in this browser.
"The DoggyFeed server did not answer within 30 seconds"	A proxy or network between you and the server holds back the save	An administrator runs the **Connection test** through the same address (section 9.5). Behind nginx, allow bodies of at least 20 MB (`client_max_body_size 20m;`).
Banner "Restart the DoggyFeed server"	The server files were updated but the server was not restarted	Whoever runs the server restarts it.
Web search says the foods come from Open Pet Food Facts	No web search key is set on the server	Add `TINYFISH_API_KEY=<key>` to the server's env file (`/etc/doggyfeed/doggyfeed.env`) and restart. Keys come from agent.tinyfish.ai.
Web search results have missing values (–)	The shop page does not list them	Choose a result with more values, or type the values from the package.
No system notifications for reminders	Notifications are off, or DoggyFeed is not opened over https	Press **Turn on notifications** in the Dog's calendar and allow them in the browser. Reminders always show in the ribbon while DoggyFeed is open.
Sign-in says to wait and try again	Too many failed sign-ins from your address or for your username	Wait the time shown, then try again with the right password.
Forgotten password		Another user of your dog house sets a new one with **Set password** on the Dog house page, or an administrator does it on the Admin page.
Buttons are missing and fields cannot be changed	You are signed in as a guest	Ask a user of the dog house to press **Allow changes** for you.
A food is missing from "Choose from food library"	Only Food library foods are offered	Add it to the Food library first (section 6).
A text is in English although another language is chosen	That text has no translation yet	It is translated in a later version; the function works the same.

Can I use DoggyFeed on my phone? Yes. The page adapts to small screens: the menu becomes a row of tabs and the bowls are shown one under another.

Can two people feed the dogs from two phones? Yes. Both sign in to the same dog house. Each page loads the other's changes within 30 seconds, or at once when it is opened again.

Does deleting a food change my history? No. Served bowls keep their own copy of each food's values.

Is DoggyFeed free? Yes. It is open source under the MIT License: free to copy, use, change and share, as long as the copyright notice naming the original author, rdone2018, is kept. Source code: github.com/rdone2018/DoggyFeed.